

Impact of Local Tumor Relapse on Patient Survival after Cobalt 60 Plaque Radiotherapy

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Abstract: The authors investigated the impact of local intraocular tumor relapse on survival in a matched-group comparison study of patients with primary choroidal or ciliary body melanoma managed with cobalt 60 plaque radiotherapy. Sixty-two patients with local relapse were matched with an equal number of relapse-free patients in terms of known clinical prognostic factors for both melanoma-specific mortality (largest linear tumor dimension, location of anterior tumor margin, age) and local tumor relapse (location of posterior tumor margin). The follow-up of every relapse-free patient equaled or exceeded the interval to relapse for each matched patient with local relapse. The estimated 5-year survival (Kaplan-Meier) in the relapse-free patients was 87% (standard error = 4%), while that in the local relapse group was 58% (standard error = 6%). This difference is statistically significant ($P < 0.0001$, log rank test). These results support the hypothesis that local tumor relapse after cobalt 60 plaque radiotherapy is an important post-treatment clinical indicator of the tumor's greater malignant potential and the patient's increased risk of melanoma-specific mortality. *Ophthalmology* 1991; 98:984-988

In a recent publication, Karlsson and co-workers¹ reported a 12% actuarial 5-year intraocular tumor relapse

rate in a group of patients with primary melanoma of the choroid or ciliary body treated with cobalt 60 plaque radiotherapy. These investigators identified the largest linear tumor dimension and the proximity of the posterior margin of the tumor to the optic nerve head as significant predictors of postplaque relapse. The authors also noted that the 5-year survival rate among patients who developed local relapse was substantially worse than that of patients who remained relapse-free. In the analysis of these relative survival rates, Karlsson and co-workers¹ did not control for intergroup differences in baseline clinical variable, which are recognized prognostic indicators of increased melanoma-specific mortality.²⁻⁶

In an effort to evaluate the impact of local tumor relapse on survival while avoiding the shortcomings of the prior investigation, we performed a matched-group comparison survival study of patients with primary choroidal melanoma who developed local relapse after cobalt 60 plaque radiotherapy versus those who remained relapse-free.

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